

Problem Set 1

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1. Within Lecture 12, we derived the following equation

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0;$$

for an elastic body. Here ρ is the referential density, $v_i = \frac{\partial \varphi_i}{\partial t}$ the velocity with φ the motion, and $\mathbf{T}$ the first Piola-Kirchhoff stress tensor.

- (a) For an arbitrary sub-body, U , of the reference body, M , show that the equality

$$\frac{d}{dt} \int_U \rho v_i d^3 \mathbf{x} = \int_{\partial U} T_{ij} \hat{n}_j dS,$$

holds, where ∂U is the boundary of U . What is the physical significance of this result? In particular, what does the vector $T_{ij} \hat{n}_j$ represent?

- (b) The angular momentum of the sub-body U about the origin is given by

$$\int_U \rho \epsilon_{ijk} \varphi_j v_k d^3 \mathbf{x}.$$

Show that a necessary and sufficient condition for this quantity to be conserved for any such U is that $\mathbf{F}\mathbf{T}^T$ is a symmetric tensor. Defining the second Piola-Kirchhoff stress, $\mathbf{S}$, through $\mathbf{T} = \mathbf{F}\mathbf{S}$, conclude that $\mathbf{S}$ is symmetric.

2. The Lagrangian density for the linearised motion for an elastic body is given by

$$\mathcal{L} = \frac{1}{2} \rho \frac{\partial u_i}{\partial t} \frac{\partial u_i}{\partial t} - \frac{1}{2} A_{ijkl} \frac{\partial u_i}{\partial x_j} \frac{\partial u_k}{\partial x_l} + \bar{S}_{ij} \frac{\partial u_i}{\partial x_j},$$

where $\mathbf{u}$ is the displacement vector, A_{ijkl} the elastic tensor, and $\bar{\mathbf{S}}$ the stress glut. Using this expression, write down the equations of motion and natural boundary conditions; here you may assume the standard Euler Lagrange equations.

The energy density for this system is defined by

$$E = \frac{1}{2} \rho \frac{\partial u_i}{\partial t} \frac{\partial u_i}{\partial t} + \frac{1}{2} A_{ijkl} \frac{\partial u_i}{\partial x_j} \frac{\partial u_k}{\partial x_l}.$$

In the absence of a stress glut, show the following equality holds

$$\frac{\partial E}{\partial t} + \frac{\partial s_i}{\partial x_i} = 0.$$

Here s_i is known as the elastic Poynting vector; the form and physical significance of this vector should be determined as part of your solution.

3. The principle of material frame indifference shows that the strain energy function, W , of an elastic body depends on the deformation gradient, $\mathbf{F}$, only through the right Cauchy-Green deformation tensor, $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and hence we can write $W(\mathbf{x}, \mathbf{F}) = U(\mathbf{x}, \mathbf{C})$, for some auxiliary function, U . The elastic tensor, A_{ijkl} is defined by

$$A_{ijkl} = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}},$$

where the partial derivative on the right hand side is evaluated at the equilibrium value $\mathbf{F} = \mathbf{I}$. Assuming that the equilibrium is stress-free, show that the elastic tensor possesses the symmetries $A_{ijkl} = A_{jikl} = A_{ijlk} = A_{klij}$. How many independent components does such an elastic tensor have? Do these symmetries remain if there is a non-zero equilibrium stress?

4. Consider plane wave propagation in a homogeneous and transversely isotropic whole space. For such a material the elastic tensor takes the form

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{kj}) + 8\gamma \hat{v}_i \hat{v}_j \hat{v}_k \hat{v}_l + 4\zeta (\hat{v}_i \hat{v}_j \delta_{kl} + \delta_{ij} \hat{v}_k \hat{v}_l) - \zeta (\hat{v}_i \hat{v}_k \delta_{jl} + \hat{v}_j \hat{v}_k \delta_{il} + \hat{v}_j \hat{v}_l \delta_{ik} + \hat{v}_i \hat{v}_l \delta_{jk}),$$

where λ , μ , γ , ζ , and ζ are elastic constants, and $\hat{\mathbf{v}}$ is a unit vector along the symmetry axis. Using first-order perturbation theory, obtain an expression for the phase speed of a quasi P-wave as a function of the angle, θ , between the propagation direction, $\hat{\mathbf{p}}$, and the symmetry axis.

5. Show that the time-dependent Schrödinger equation for a free particle

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi,$$

admits plane wave solutions of the form

$$\psi(\mathbf{x}, t) = a \exp \left[\frac{-i(Et - \mathbf{p} \cdot \mathbf{x})}{\hbar} \right]$$

and obtain the relationship between the energy E and momentum $\mathbf{p}$. Consider the corresponding equation for a particle in a non-constant but smoothly varying potential field V :

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi.$$

Starting from the ansatz

$$\psi(\mathbf{x}, t) = a(\mathbf{x}) \exp \left\{ \frac{-i[Et - \varphi(\mathbf{x})]}{\hbar} \right\},$$

where the amplitude a and phase function φ are to be determined, and retaining only the leading-order terms in powers of $1/\hbar$, obtain the eikonal equation

$$H(\mathbf{x}, \nabla \varphi) = E$$

where H is the classical Hamiltonian. Applying the method of characteristics, show that the particle's phase function can be determined through solution of the classical equations of motion.

6. Consider P-wave rays within an isotropic body. The travel-time of a ray between two points $\mathbf{x}_1$ and $\mathbf{x}_2$ can be written

$$T = \int_0^1 \frac{1}{\alpha(\mathbf{x})} \sqrt{\frac{dx_i}{d\gamma} \frac{dx_i}{d\gamma}} d\gamma,$$

where the $\gamma \mapsto \mathbf{x}(\gamma)$ is the ray path between $\mathbf{x}_1$ and $\mathbf{x}_2$ parametrised by a generating parameter $\gamma \in [0, 1]$. Regarding this travel time as a functional of the given ray path, Fermat's principle states that the true ray path is a stationary point of this functional. Obtain the corresponding Euler-Lagrange equations, and show that they are equivalent to the Hamiltonian ray tracing equations discussed within lectures. [Hint: it will be useful to express both sets of ordinary differential equations in terms of the arc length along the ray.]

7. Consider an isotropic elastic half-space in which the material parameters vary only with depth $z \geq 0$. Suppose that the P-wave speed α increases monotonically, and consider a ray starting at the surface whose initial tangent vector makes an angle $0 < \theta_0 < \pi/2$ with the z -axis and lies in the $x - z$ plane. Assume that the ray travels monotonically down into the half-space until a depth $z_t > 0$ at which point it is traveling horizontally, and subsequently it turns upwards, returning to

the surface in a symmetric manner. By parametrising this ray in terms of the depth co-ordinate z , show that its travel time can be written

$$T = \int_0^{z_t} \frac{2}{\alpha} \sqrt{1 + \frac{dx_i}{dz} \frac{dx_i}{dz}} dz,$$

where $\mathbf{x}$ denotes the horizontal position vector of the ray given as a function of z . Apply Fermat's principle to show that the ray remains in its initial plane, and that along the ray path

$$\frac{\sin \theta}{\alpha} = q$$

where θ denotes the local angle between the vertical direction and the tangent to the ray, and q is a constant known as the ray parameter. Obtain, in particular, an implicit equation for the turning depth z_t in terms of q . Show that the travel time T and total horizontal distance X of the ray can be obtained through the following integrals

$$T = \int_0^{z_t} \frac{2}{\alpha} \frac{dz}{\sqrt{1 - q^2 \alpha^2}}, \quad X = \int_0^{z_t} \frac{2q\alpha}{\sqrt{1 - q^2 \alpha^2}} dz.$$